

HOLLAND S. HARGENS

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Computer Engineer with hands-on experience in ML model development, scientific data pipelines, and containerized software. Built and benchmarked models on real sensor data, comfortable in Python, Linux, and Docker from internship and project work. Seeking roles in software, signal/data processing, or applied ML.

EDUCATION

University of Wisconsin–Madison | May 2026

Bachelor of Science in Computer Engineering | GPA: 3.6/4.0 | Dean's Honor List (Fall 2022 – Spring 2026)

Relevant Coursework: Software Engineering · Database Management · Computer Architecture · Microprocessor Systems · Digital System Design · Electronic Circuits · Machine Organization & Programming · Artificial Intelligence

EXPERIENCE

Software & AI Engineer | [Radius Hire](#) · Remote (Feb 2026 – Present)

- ▶ Engineered an automated interview scheduling flow using FastAPI and the WhatsApp Business API, eliminating manual recruiter coordination across daily candidate sessions
- ▶ Implemented an AI-driven video screening pipeline with client-side compression and cloud storage integration, minimizing load time and storage overhead
- ▶ Maintained and debugged a Gemini API-integrated candidate screening workflow across a Node.js/TypeScript + Next.js platform, resolving integration issues and managing in a Docker-containerized environment

ML & Software Engineering Intern | [Veridis Technologies](#) · Amsterdam, Netherlands (Jun – Aug 2025)

- ▶ Designed a preprocessing pipeline for 23-channel time-series polymer sensor data, engineering feature extraction, and normalization across validation splits to ensure reliable downstream model training on scientific instrumentation data
- ▶ Trained and benchmarked regression and classification models in PyTorch and TensorFlow against real sensor measurements, achieving ~7% RMSE on polymer blend estimation across 50 cross-validated samples
- ▶ Systematically evaluated 20+ experiment configurations in MLflow, sweeping learning rate, batch size, and regularization hyperparameters to identify best-performing architectures across a scientific dataset

PROJECTS

PSoC6 Blackjack — Bare-Metal Embedded System — C · ARM Cortex-M4 · SPI · UART · I2C · GPIO

- ▶ Implemented a complete Blackjack game in embedded C on a PSoC6 ARM Cortex-M4, integrating hardware timers, GPIO button handling, and interrupt-driven input processing with debounce logic
- ▶ Drove an LCD over SPI, managing frame updates and state rendering across 6 game states within timing constraints
- ▶ Integrated 4 peripherals sharing interrupt logic, validating correct state-machine transitions across all game scenarios

FPGA Capstone — On-Device ML Workout Detection — Xilinx Zynq · PS↔PL · Verilog · Python · MLP

- ▶ Implemented PS-to-PL communication on a Xilinx Zynq SoC to synchronize data transfer between an ARM core and FPGA fabric for low-latency on-device ML inference
- ▶ Trained a multi-layer perceptron to classify 6 workout exercises across 5 test subjects, achieving ~90% accuracy at <15ms inference latency with fully on-device operation
- ▶ Managed the full signal acquisition and processing pipeline from camera module input through on-device MLP inference to LCD output, validating end-to-end latency under real operating conditions

Goal Planner App — AI-Powered Full-Stack Platform — Node.js · Next.js · TypeScript · AWS · Docker · MySQL

- ▶ Architected a full-stack TypeScript/Next.js + MySQL application containerized with Docker for consistent staging and production deployment on AWS (EC2, S3, RDS), and integrated with Anthropic API
- ▶ Built an adaptive weekly AI plan regeneration pulling prior task completions to generate an upcoming plan

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, TypeScript, Verilog, SystemVerilog, MATLAB

Embedded & Hardware: ARM Cortex-M4, PSoC6, UART, SPI, I2C, GPIO, RTOS, Xilinx Zynq/FPGA, Altium PCB Design

Web & Backend: Node.js, Next.js, React, FastAPI, REST APIs, Java Spring Boot, MySQL, PostgreSQL

AI / ML: PyTorch, TensorFlow, Scikit-learn, MLflow, Pandas, Hugging Face, Gemini API

Cloud & DevOps: AWS (EC2, S3, RDS, IAM), Docker, Docker Compose, GitHub Actions, Git, Linux

ACTIVITIES & LEADERSHIP

Student Member | [IEEE UW-Madison Chapter](#) (2023 – 2026)

- ▶ Contributed to a weather balloon project focused on embedded sensor integration and high-altitude data collection

Eagle Scout | Troop Patrol Leader | [BSA](#) (2018 – 2019)

- ▶ Led a patrol of 20 scouts, planned and executed community service projects, and mentored junior members